

"The James Webb Space Telescope (JWST), launched on December 25, 2021, is the premier observatory of the next decade, serving thousands of astronomers worldwide. It orbits the Sun at the second Lagrange point (L2), approximately 1.5 million kilometers from Earth. Unlike its predecessor, the Hubble Space Telescope, which observes primarily in the near-ultraviolet, visible, and near-infrared spectra, JWST is optimized for infrared astronomy. This allows it to peer through dense dust clouds and observe some of the oldest, most distant galaxies in the universe. The telescope's primary mirror consists of 18 hexagonal beryllium segments coated in gold, giving it a total diameter of 6.5 meters. A critical component of JWST is its five-layer sunshield, made of Kapton, which keeps the telescope's instruments below 50 Kelvin (-223°C) to prevent the telescope's own infrared emissions from interfering with astronomical signals.